

# List of Figures



|      |                                                                                                                                                               |    |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1.1  | The locations of the $\sim 9000$ stars in the night sky visible to the naked eye. . . . .                                                                     | 4  |
| 2.1  | A schematic of the reactions leading up to the CNO cycle. . . . .                                                                                             | 17 |
| 2.2  | The evolution of an $4 M_{\odot}$ star from ZAMS until CO core development. . . . .                                                                           | 20 |
| 2.3  | The evolution of stars in the range $0.5 - 80 M_{\odot}$ . . . . .                                                                                            | 21 |
| 2.4  | The evolution of an $8 M_{\odot}$ star. . . . .                                                                                                               | 22 |
| 2.5  | A rough outline of supernova classification. . . . .                                                                                                          | 24 |
| 2.6  | The binding energy per nucleon and half-lives for elements up to Oganesson. . . . .                                                                           | 25 |
| 2.7  | The HRD for an $21 M_{\odot}$ star with an (unshown) $10 M_{\odot}$ companion with varying period. . . . .                                                    | 26 |
| 2.8  | A schematic of a simple binary star system. . . . .                                                                                                           | 28 |
| 2.9  | The dimensionless Roche potential as a function of mass ratio $q$ . . . . .                                                                                   | 29 |
| 2.10 | A schematic of the Roche lobe formula Eqn(2.15). . . . .                                                                                                      | 30 |
| 2.11 | The volume of the Roche Lobe computed numerically through Monte-Carlo integration vs the<br>volume assuming Eqn(2.15). . . . .                                | 30 |
| 2.12 | As for Figure 2.11, but for the $\mathcal{L}_2$ point. . . . .                                                                                                | 31 |
| 2.13 | Two stars of a similar mass that undergo a helium flash and two thermal pulses. . . . .                                                                       | 42 |
| 2.14 | The helium flash shown on a Kippenhahn diagram. . . . .                                                                                                       | 43 |
| 3.1  | The value of the meshpoint sum $\mathcal{C}$ in Eqn(3.2) throughout VCA. . . . .                                                                              | 51 |
| 3.2  | The main evolution of a binary with VCA enabled. . . . .                                                                                                      | 53 |
| 3.3  | As for Figure 3.2, but for the NVCA model. . . . .                                                                                                            | 54 |
| 3.4  | The evolution of the radius, rotation rate, and luminosity for the VCA and NVCA models,<br>shown around the contact phase. . . . .                            | 55 |
| 3.5  | The Kelvin-Helmholtz time-scales of both stars in the VCA and NVCA regime. . . . .                                                                            | 57 |
| 3.6  | A Hertzsprung-Russell Diagram showing the evolution of our model both with and without the<br>VCA prescription. . . . .                                       | 58 |
| 3.7  | The abundance of helium at the five critical stages of the binary evolution. . . . .                                                                          | 60 |
| 3.8  | The helium abundance in the accretor near when the model detaches. . . . .                                                                                    | 60 |
| 3.9  | An Hertzsprung-Russell Diagram of the post-mass transfer secondary model evolved as a single<br>star with different thermohaline mixing coefficients. . . . . | 61 |

|      |                                                                                                                                                                                                                                  |     |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3.10 | The helium abundance profile for each model of thermohaline mixing, as a function of the internal mass of the star. . . . .                                                                                                      | 64  |
| 3.11 | The (evolutionary) mass-luminosity relationship in the presence and absence of VCA. . . . .                                                                                                                                      | 66  |
| 4.1  | The PDF and CDF of the distributions using kicks from <a href="#">Hobbs et al. (2005)</a> ; <a href="#">Verbunt et al. (2017)</a> ; <a href="#">Bray &amp; Eldridge (2018)</a> ; <a href="#">Willcox et al. (2021)</a> . . . . . | 72  |
| 4.2  | A schematic of how FINEGRID and WIDEGRID are related. . . . .                                                                                                                                                                    | 73  |
| 4.3  | The effect of varying $d_e$ in Eqn(4.5) on the coalescence time. . . . .                                                                                                                                                         | 76  |
| 4.4  | The effect of varying $d_e$ in Eqn(4.5) on the runtime and performance. . . . .                                                                                                                                                  | 76  |
| 4.5  | The merger rate as a function of kick parameters $\alpha$ and $\beta$ . . . . .                                                                                                                                                  | 78  |
| 4.6  | The period-eccentricity distributions for six selected <a href="#">Bray &amp; Eldridge (2016)</a> kicks. . . . .                                                                                                                 | 79  |
| 4.7  | The $\ln \mathcal{K}$ values across $\alpha - \beta$ space. . . . .                                                                                                                                                              | 81  |
| 4.8  | The KS statistic for FINEGRID and WIDEGRID. . . . .                                                                                                                                                                              | 82  |
| 4.9  | The PDFs and CDFs of four systems, alongside the Hobbs CDF and <a href="#">Willcox et al. (2021)</a> CDF. . . . .                                                                                                                | 83  |
| 4.10 | The four constraints we apply overlaid on our FINEGRID parameter space. . . . .                                                                                                                                                  | 85  |
| 4.11 | The CDF for a population of isolated pulsars and the period-eccentricity distribution for $\alpha = 115 \text{ km s}^{-1}, \beta = 15 \text{ km s}^{-1}$ . . . . .                                                               | 86  |
| 4.12 | The peak of the event rate $\mathcal{R}(t)$ as a function of $\alpha$ and $\beta$ . . . . .                                                                                                                                      | 87  |
| 4.13 | The event rate $\mathcal{R}(z)$ for a region surrounding $\alpha = 330, \beta = -70$ on FINEGRID. . . . .                                                                                                                        | 88  |
| 5.1  | BPASS v2.2.1 primary models for the matches obtained by HOKI for the parameters in Table 5.1. . . . .                                                                                                                            | 101 |
| 5.2  | BPASS v2 model tracks for our six close-contact binaries. . . . .                                                                                                                                                                | 102 |
| 5.3  | Detailed model tracks for our six close-contact binaries. . . . .                                                                                                                                                                | 106 |
| 5.4  | The mass-radius plots for our models. . . . .                                                                                                                                                                                    | 108 |
| 5.5  | HRDs for the matching overcontact binary models. . . . .                                                                                                                                                                         | 109 |
| 5.6  | The three fits for TU Muscae we recover. . . . .                                                                                                                                                                                 | 112 |
| 5.7  | The dependence of the mean contact length on the initial parameters $\log P$ , $M_1$ , and $q$ . . . . .                                                                                                                         | 114 |
| 6.1  | The geometry of an ellipse and the circle with a radius equal to the semi-latus rectum. . . . .                                                                                                                                  | 120 |
| A.1  | A schematic of a simple binary star system. . . . .                                                                                                                                                                              | 128 |
| A.2  | An infinitesimal area segment of an orbital trajectory. . . . .                                                                                                                                                                  | 131 |
| A.3  | The Chebyshev polynomial expansion of Eqn(A.92). . . . .                                                                                                                                                                         | 139 |
| A.4  | An illustration of the bi-stability jumps in the mass loss prescription. . . . .                                                                                                                                                 | 142 |
| C.1  | The period-eccentricity distributions of populations governed by Bray kicks across the LIGO O3 contour. . . . .                                                                                                                  | 149 |
| D.1  | The results of our bespoke binary evolution for normal periods. . . . .                                                                                                                                                          | 157 |
| D.2  | The results of our bespoke binary evolution for wider periods. . . . .                                                                                                                                                           | 161 |

|                                                                              |     |
|------------------------------------------------------------------------------|-----|
| D.3 The results of our bespoke binary evolution for shorter periods. . . . . | 165 |
|------------------------------------------------------------------------------|-----|

